NCAN · WADSWORTH CENTER · SUMMER 2026

Can the brain tell us how big the reflex will be?

Machine learning on 2.7 million trials of twenty-year-old neural recordings.

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with Dr. Jonathan Carp and Teresa

WHAT WE'LL COVER

Roadmap

1	**The question**	what conditioning is, and what was already known
2	**The data**	one trial, six animals, and a broken decoder
3	**How we built it**	the model, the stack, the daily workflow
4	**What we found**	two results, one negative, and the controls
5	**Where it goes**	what we'd publish, and what it's useful for

What the animals are doing

1

The reflex

A small shock to the leg nerve produces a reflex in the soleus muscle. Its size is what we measure.

2

The training

The rat is rewarded whenever that reflex is bigger than a threshold — or smaller, depending on the group.

3

The learning

Over weeks the reflex shifts in the trained direction. A 20% change counts as success, and the change is in the spinal cord.

Already known: cortical activity influences reflex size (Boulay, Chen & Wolpaw, 2015). **Our question:** is it visible on a single trial, and does it change as the animal learns?

Why bring machine learning to this

The usual approach

Decide in advance which features of the brain trace might matter — a peak here, a band of power there — then measure those.

Whatever you didn't think of, you don't measure.

What we did instead

Let a network learn its own compact description of each trace, with no labels and no assumptions, then test what that description can predict.

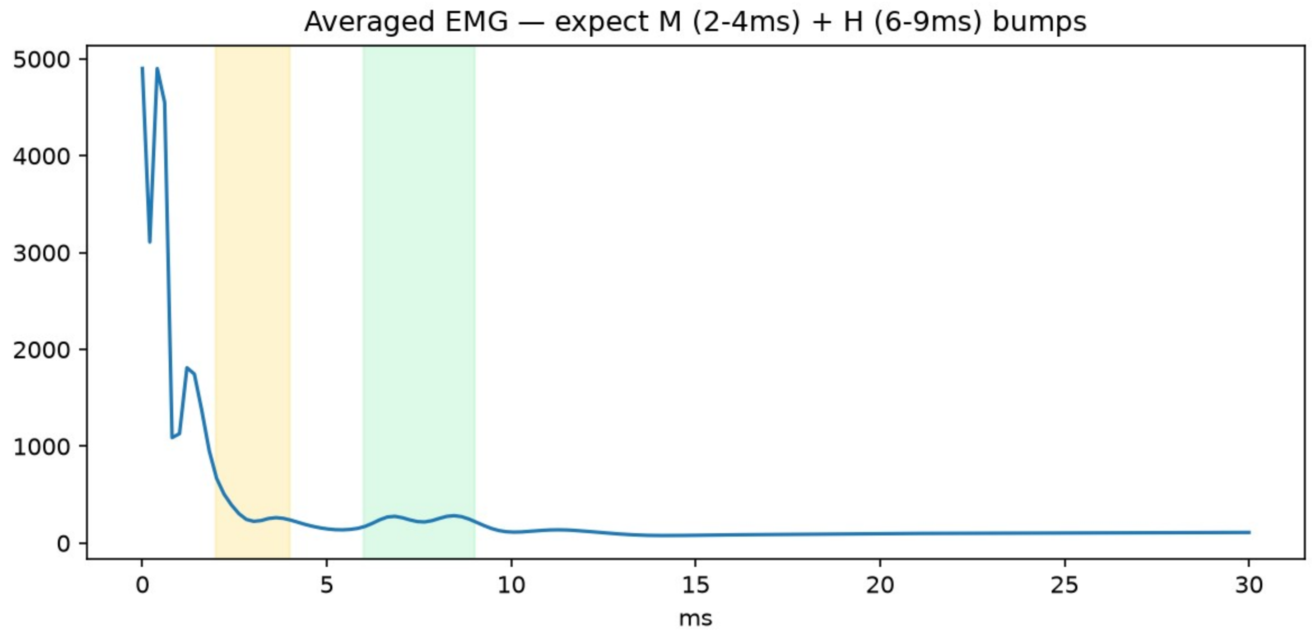
With 2.7 million trials, that is tractable.

The precedent

This mirrors what BrainBERT did for human intracranial recordings — learn a general representation of neural signal first, without labels, then use it for whatever downstream question you have. We use a much smaller model, because we have one channel and a 30 ms window rather than a whole electrode array.

What a single trial looks like

Every shock produces one trial: a 30-millisecond window, two electrodes recorded at the same moment.



Reading the trace

- **The spike at zero**

  the stimulus itself — an electrical artifact

- **M-wave, 2–4 ms**

  the nerve firing the muscle directly

- **H-reflex, 6–9 ms**

  the signal that travelled to the spinal cord and back — this is what conditioning changes

First problem: the archive would not decode

What we were given

2006 MySQL tables and a decoder script. The script documented the byte order as big-endian.

What happened

Decoded as documented, every trial came out as noise. No M-wave, no reflex, nothing.

What was wrong

The files are little-endian. One flag. Flip it and the physiology appears exactly where the experimenters' own notes said it would.

How we knew we had it right

Not because it looked plausible — because the M-wave and H-reflex landed at exactly the latencies the experimenter had typed into the log in 2006: “MR interval 2-4; HR interval 6-9”. The data confirmed the notes and the notes confirmed the data.

Six animals, 2.7 million trials

2.7M	**6**	**3 / 3**	**45–210**
trials decoded	animals	down / up	days each

Choosing which animals to download, without downloading them

The log files are a few hundred kilobytes; the signal data is four gigabytes an animal. The experimenter raises the reward threshold whenever the rat starts beating it, so the threshold history is a written record of how the animal was doing. We read only the logs, then downloaded signal data for the animals that had actually learned.

One animal excluded. A12's pre-stimulus cortical signal is 2.8 µV against 106–128 µV in the others — that electrode was dead.

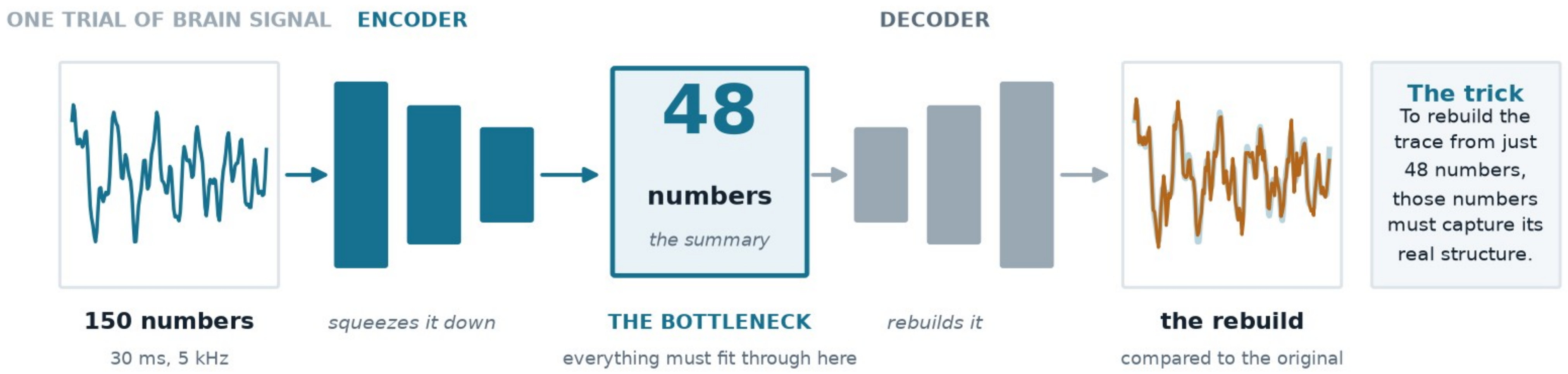
The pipeline, end to end

01	02	03	04	05	06
Archive	**Decode**	**Store**	**Measure**	**Learn**	**Test**
2006 MyISAM tables	little-endian int16 → µV	one Zarr per animal	M-wave, H-reflex, background	48-number summary	against five controls
`MySQL`	`NumPy`	`Zarr`	`SciPy`	`PyTorch`	`scikit-learn`

2.7 million trials · 6 animals · about 4 GB in per animal, 300 MB out · two commands per animal, start to finish

Everything scripted and version-controlled. Re-running an animal from scratch takes minutes, not a day of clicking.

What the model actually is



Nobody tells it what to look for.

We never show it an H-reflex, a label, or a training phase. It only ever tries to rebuild what it was given — so the 48 numbers are whatever the brain signal's own structure demands, rather than the features we assumed mattered.

Training it, and the one trap we avoided

How it was trained

600,000	trials pooled across five animals
12	passes through the data
~4 min	on a laptop GPU
0.0149 → 0.0070	reconstruction error

The trap

Pool several animals naively and the model learns to tell the animals apart — different electrodes, different amplitudes. That is a recording artefact wearing a biology costume.

Fix: standardise every animal separately before pooling.

Why one model across all animals, rather than one per animal

A separate model per animal would describe each brain in its own private vocabulary, and you could never compare them. Training one model on all of them forces a shared vocabulary — so "this pattern of cortical activity" means the same thing in animal 3 as it does in animal 11. That is what makes the group comparisons later in the talk legitimate.

Tech stack and daily workflow

DATA	MODEL	ANALYSIS	DELIVERY
• MySQL / MariaDB	• PyTorch	• scikit-learn	• Git / GitHub
• Zarr	• Apple MPS / GPU	• SciPy	• matplotlib
• NumPy	• 48-dim encoder	• ridge + permutation	• automated figures

Day to day: read the log → load one animal → convert → eyeball the averaged trace → run the analyses → commit figures.

Every figure in this talk regenerates from a single command. Nothing was hand-edited.

Why reflex size alone tells you nothing

The problem

A bigger shock makes a bigger reflex — no learning required. Across these recordings the delivered stimulus roughly quadrupled.

Early on this made one of our results come out backwards.

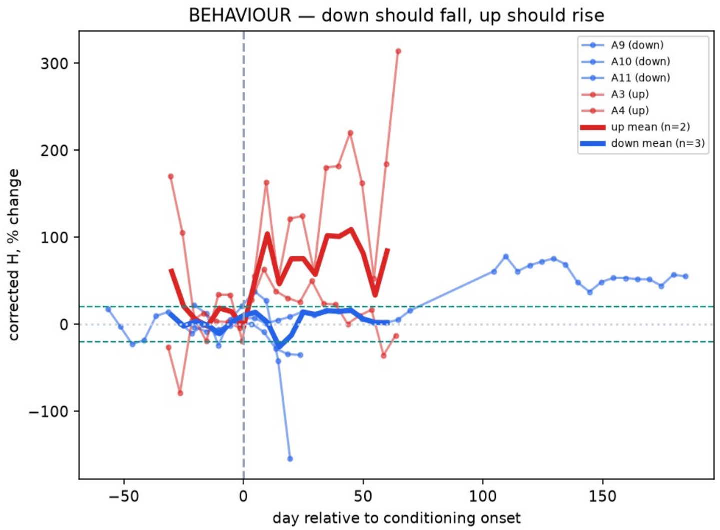
The fix

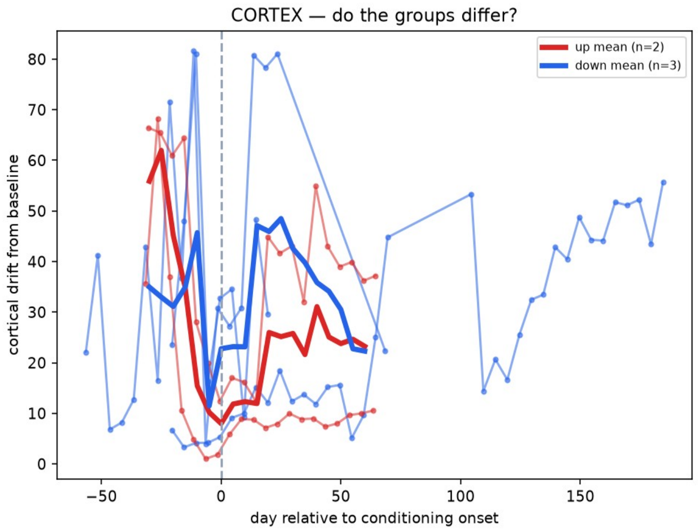
The M-wave sits in the same trace and reflects only the stimulus. It is a per-trial receipt for how much shock arrived. We remove its influence from every measure.

Dr. Carp's addition: the reflex also depends on how active the muscle already was. We now measure the 20 ms before each shock and remove that too.

Every number in the rest of this talk has both corrections applied.

Conditioning works, in both directions





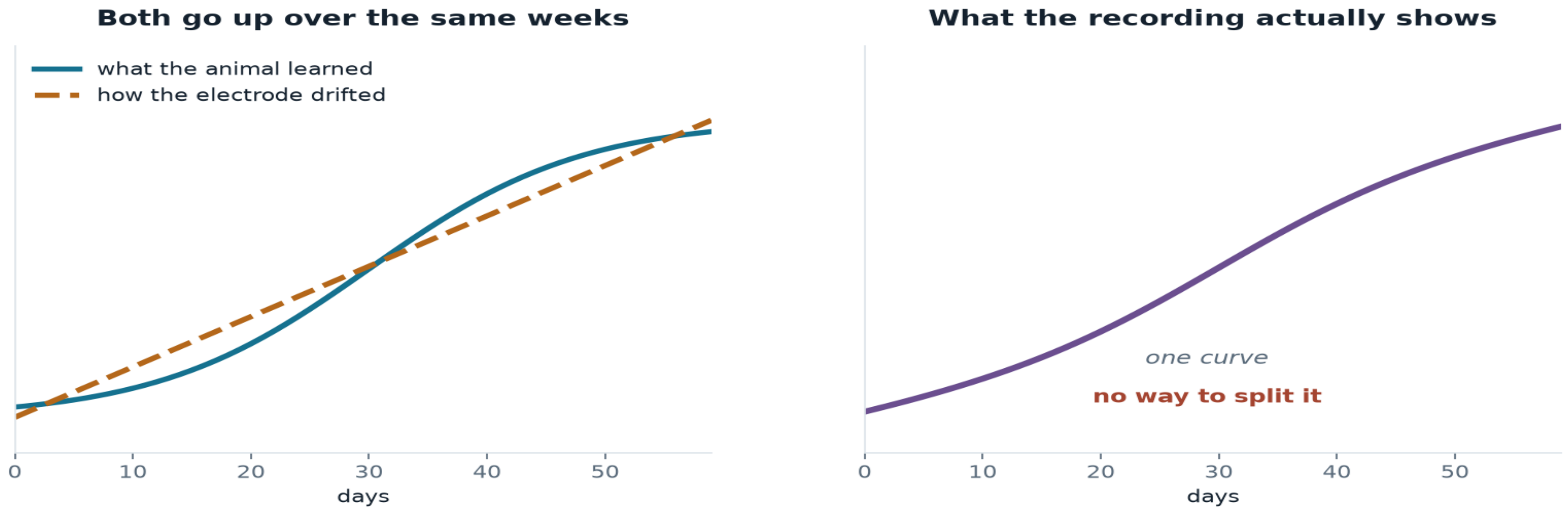
The positive control

Each animal is lined up to its own training start day. Up-trained animals (red) rise; down-trained (blue) fall.

Three of five clear the 20% criterion in the trained direction.

The two that don't become a built-in control later on.

Two things change slowly over the same weeks



Inside a single animal, learning and electrode drift are both smooth functions of time. That is why our first analysis could not work.

This is a limit of the study design, not of the data — and it shaped everything we did next.

PART 4 · RESULT TWO, A NEGATIVE

Our first approach did not survive its controls

We measured how far the average brain state moved across training. It moves — but three tests say the movement is the recording, not the animal.

Take baseline-only data and pretend half was training	Nothing happened, so it should show no movement	**Moves just as much — 5 of 6 animals**
Shuffle which trial belongs to which day	Keeps the numbers, destroys the timeline	**No movement — the measure itself is sound**
Compare animals trained up against trained down	Electrode drift doesn't care which way you trained	**No difference between the groups**

Two of these three were Dr. Carp's suggestions. They changed our conclusion.

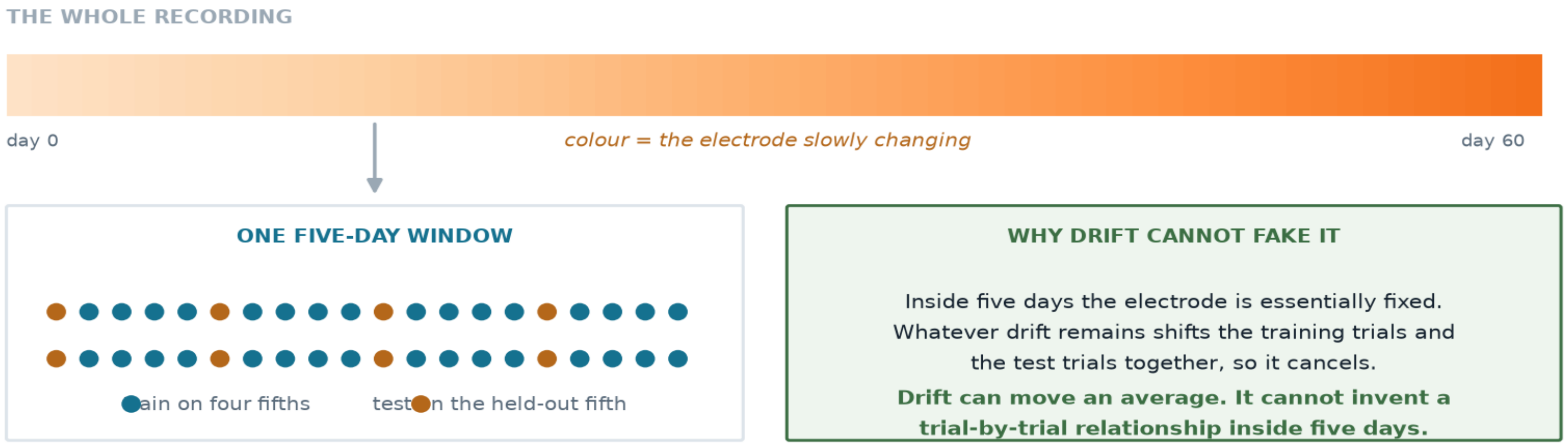
So we asked a question drift cannot fake

Instead of

“Did the average brain state move over weeks?”

We ask

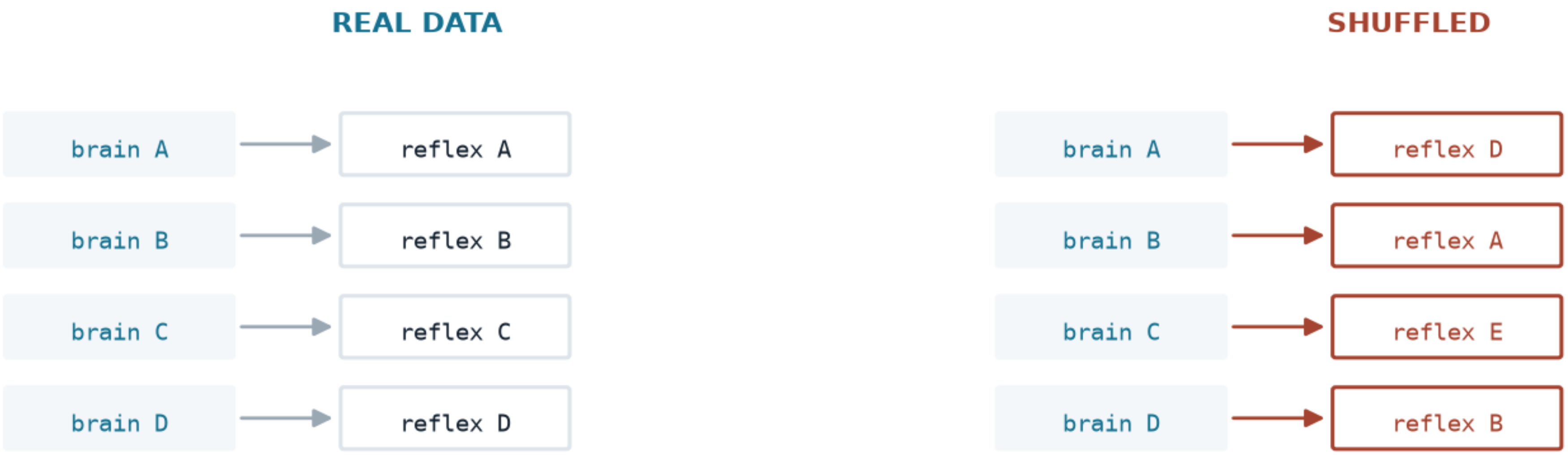
“Can the brain predict this trial's reflex?”



We repeat this in every window — 101 of them across five animals — and compare each against a shuffle of its own trials.

A window only counts if the real data beats its own shuffled twin.

And each window is judged against itself



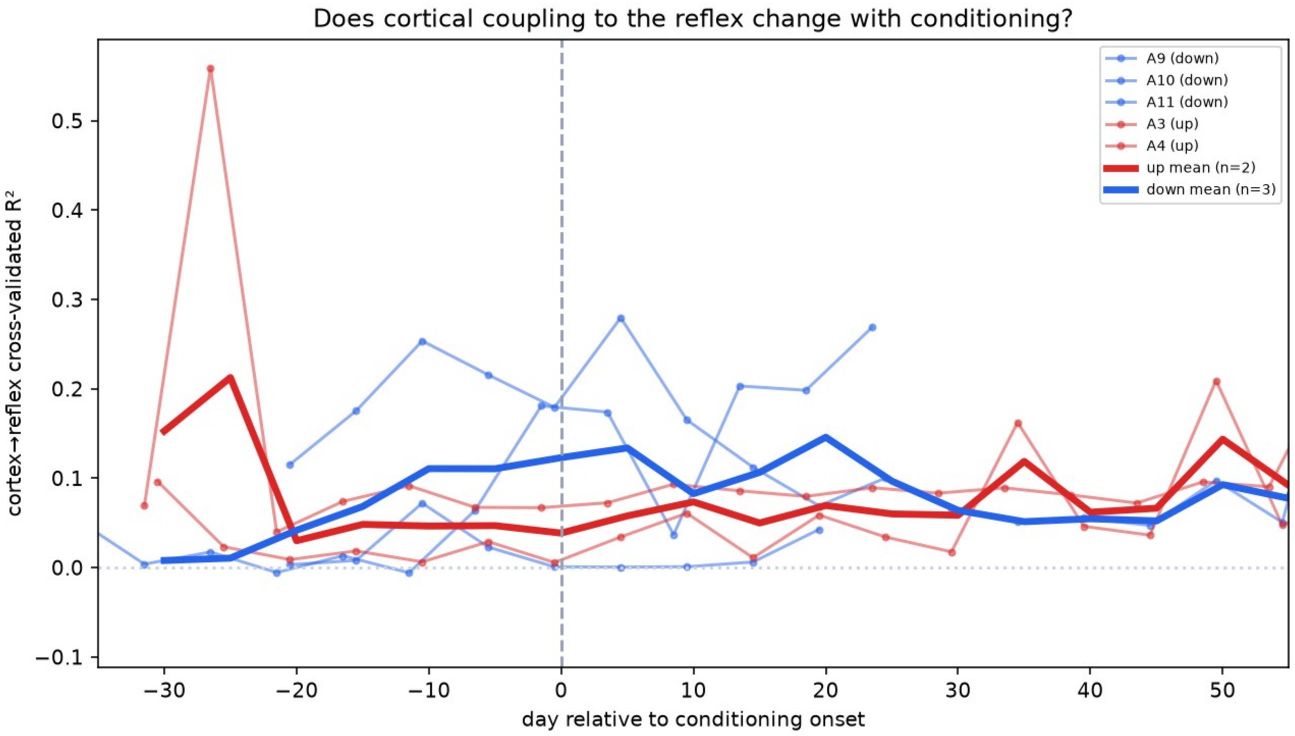
Whatever relationship exists, the model can find it.

Same numbers, wrong pairings. Any relationship is destroyed.

If the real version doesn't beat the shuffled one, there was nothing there.

We repeat this in all 101 windows across five animals. A window only counts if the real data beats its own shuffled twin.

The brain does carry a signal about the reflex



99 of 101

windows where the real data beat its own shuffled version

Present in all five animals.

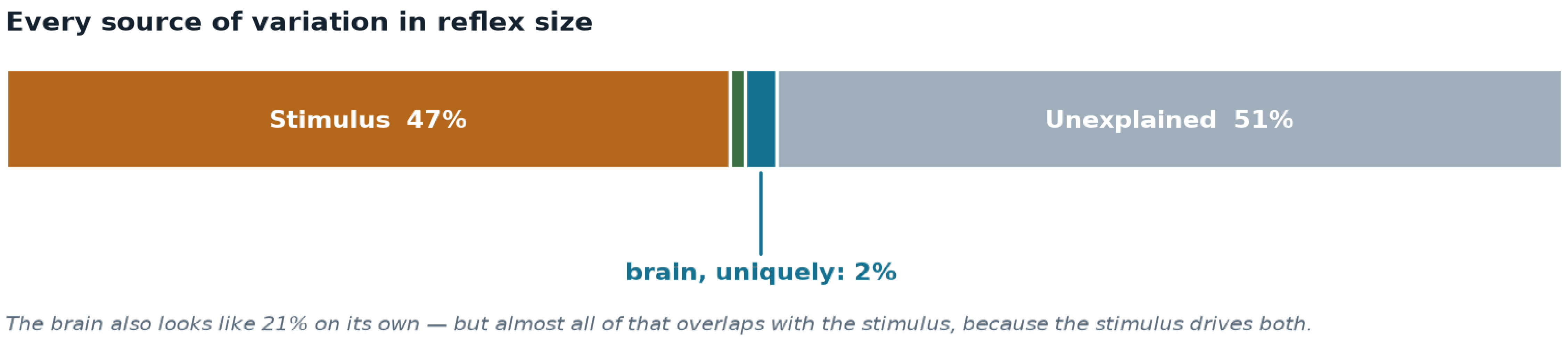
Counting the animal as the unit of analysis:

$t(4) = 3.81$, $p = 0.019$

Small in size, but almost never absent.

How much does the brain actually explain?

Dr. Carp asked us to split the variance up. The stimulus dominates, and the brain overlaps with it heavily.



Why the brain's own 21% collapses to 2%

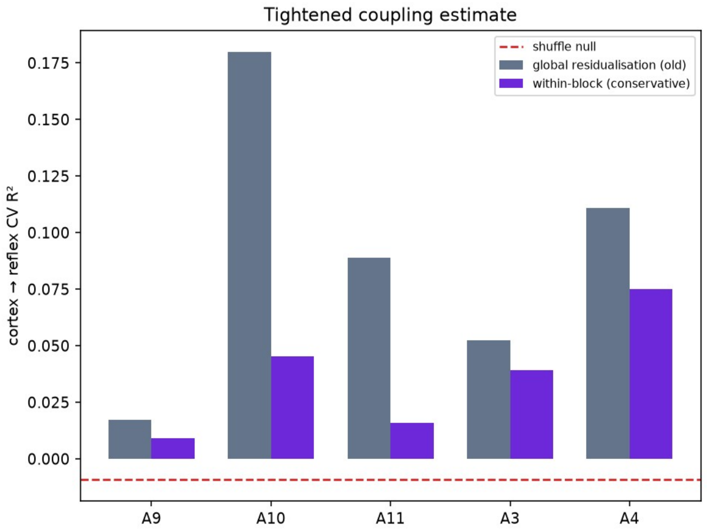
The stimulus drives both the cortical evoked response and the reflex, so most of what the brain appears to explain is the stimulus wearing a different hat. Only the part that survives once stimulus and background are already in the model counts as a genuine cortical contribution.

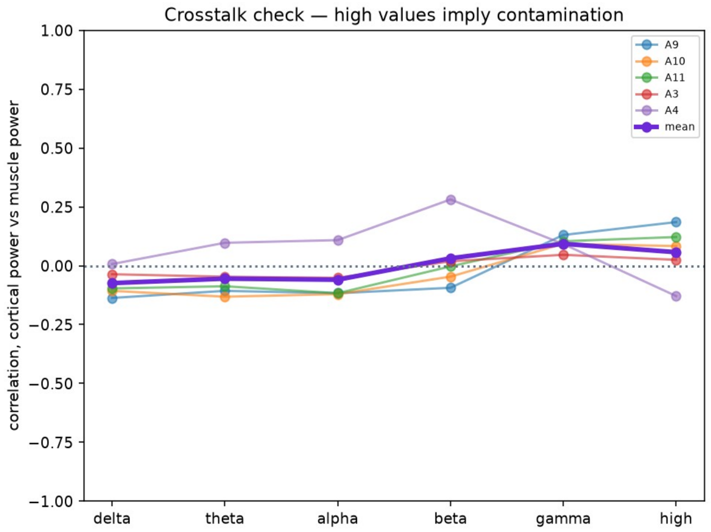
Everything we tried to kill it with

Baseline sham	Pretend part of the baseline was training	drift fails
Shuffled trial order	Keep the numbers, destroy the timeline	null ≈ 0
Stimulus strength	Is it just how hard we shocked?	survives
Pre-stimulus background	Is it muscle tone rather than brain?	survives
Within-window testing	Could electrode drift manufacture it?	drift cancels
Muscle crosstalk	Is the brain electrode picking up EMG?	r = +0.06
Up versus down	Does it depend on training direction?	no difference

Five it survives, two it fails — and the two failures are what told us the average-state approach was the wrong question.

Is the brain electrode just picking up muscle?





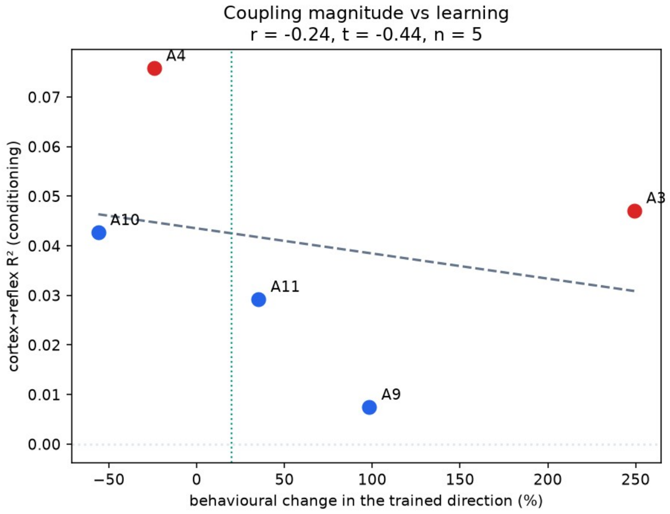
The test

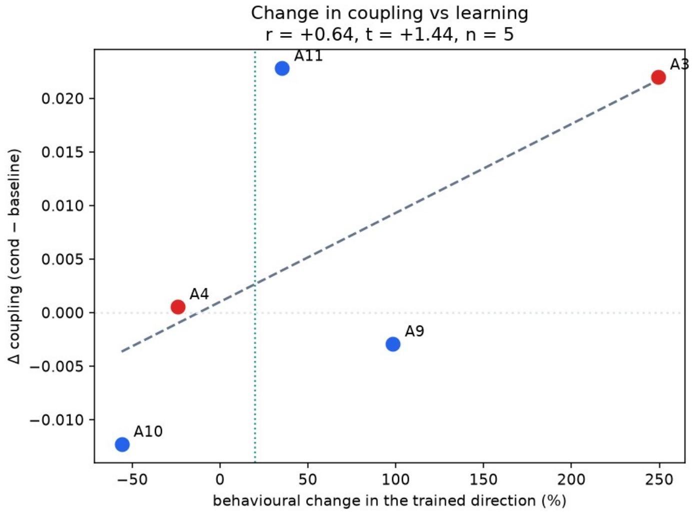
Both channels record at once, so the cortical electrode could in principle be registering muscle activity. Muscle artefact lives at high frequency, so that is where contamination would appear.

Above 100 Hz: r = +0.06

Essentially zero, and the low bands are slightly negative. The cortical signal also marginally leads the muscle, which is the acceptable direction.

Does it scale with how much they learned?





r = +0.71

between how much an animal learned and how much its cortical coupling changed

The two animals that failed to learn show no increase — a built-in negative control.

With five animals this is a direction, not a result. Five more are ready to process.

PART 5 · A NEW THREAD

The metadata may be a measurement

An idea that came out of the last meeting, and one we think is worth pursuing.

What the reward threshold is

The experimenter raises it when the rat is beating the task and lowers it when the rat is struggling. It is adjusted by hand, day by day, across the whole experiment.

It is, in effect, a staircase procedure.

Why that is interesting

It is a record of how hard the task was for that animal — its engagement and effort — written down independently of anything we measure from the electrodes.

No shared noise with our signal processing.

Right now our scaling result correlates two things measured from the same electrodes.

If cortical coupling instead tracks a behavioural record kept by hand at the time, that is a far harder result to dismiss. We would like to build this next.

Three things worth writing up

The method

A reproducible pipeline that recovers a twenty-year-old archive and learns a representation from it, on a laptop.

Useful to anyone with old chronic recordings sitting on a shelf.

The finding

Cortical activity carries a small, highly reliable single-trial signal about reflex size, independent of stimulus and excitability.

Extends Boulay et al. from a population relationship to a single-trial one.

The warning

The intuitive drift analysis produces a convincing result that three controls reject.

We would want to save the next group from spending months on it.

Target: a machine-learning venue, framed as an extension of the existing cortical–spinal work.

Where a single-trial cortical readout could help

Closing the loop

If cortex predicts the reflex before the fact, conditioning protocols could adapt trial by trial rather than day by day.

Understanding who responds

Roughly a third of animals — and people — don't respond to conditioning. A cortical readout might show why, early.

Rehabilitation

Reflex conditioning is already used clinically after spinal cord injury. Any handle on the cortical side is a handle on individualising it.

All three depend on the effect being larger and more reliable than what we can currently demonstrate. We are not there yet — but this is the direction that would make it worth getting there.

PART 5 · NEXT

What happens next

1	**Process the remaining animals**	Five are downloaded and ready. Takes the sample from five to about ten.	*this week*
2	**Power the scaling test**	With ten animals the learning-versus-coupling correlation becomes a real test.	*next*
3	**Frequency-resolved coupling**	Which bands carry the reflex information? Sharpens the finding and the contamination argument.	*next*
4	**The reward-criterion measure**	An independent behavioural record to correlate against.	*new thread*
5	**Write it up**	Method, finding and warning.	*after that*

WHERE WE'D VALUE YOUR INPUT

Four questions

1 Is a 2% unique contribution worth reporting?

Small, but consistent, and it survives every control we could think of.

2 Does the within-window design convince you?

Train and test inside five days so drift cancels. Enough to rule out the electrode?

3 What else could produce this?

We've excluded stimulus, muscle tone, drift and crosstalk. What are we missing?

4 Is the negative result worth publishing?

The obvious analysis looks convincing and fails three controls. Useful warning, or noise?

Code, figures and the full write-up: github.com/switchrao777/hroc-ncan